MILITARY SPECIFICATION

AUDIOMETER, AUTOMATIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an automatic von Bekesy type audiometer, complete with mechanical recorder, earphones, patch cords, patient operated hand switch, pens and recording charts.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-L-116	Preservation, Methods of.
MIL-M-7298	Manual, Technical: Commercial Equipment.
MIL-C-37187	Chart, Recording Instrument, Audiometer.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6515

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

Close Coupler Type 9A

(Application for copies should be addressed to Superintendent of Documents, Washington, DC 20025).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

S3.6 - 1969	Specifications for Audiometers.
C73.11 - 1966	Plugs and Receptacles - General Purpose 125 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 498	Electrical Attachment Plugs and Receptacles

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, L.I., N.Y. 11746; 207 East Ohio Street, Chicago, IL 60611, 1655 Scott Boulevard, Santa Clara, CA 95050, 333 Pfingsten Road, Northbrook, IL 60062.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout, suitable in all respects for the intended purpose.

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3.2 Compliance with American National Standards Institute Specifications. Audiometer shall be in accordance with the requirements of ANSI S3.6. Compliance with these requirements does not absolve the contractor from complete compliance with the requirements of this specification in order to secure acceptance of his commodity. In the event of differences between the requirements of this specification and other specifications referenced herein, the requirements of this specification shall govern.

3.3 Style and design. Audiometer shall be a fixed-frequency, automatic recording, von Bekesy type, solid-state construction. The subject under test shall be able to control the direction of variation in test tone amplitude. Variation in test tone amplitude shall be accomplished by means of an attenuator which is interposed between the earphones and an oscillator which generates the test tone. The oscillator shall operate to apply a signal to a ladder attenuator. The audiometer shall have the capability of automatically plotting the audiogram on an audiometer chart as controlled by the hand switch operated by the subject under test. The audiometer shall have a zero reference level and shall be suitable for determining hearing loss in decibels (db.). The audiometer shall be free from electrical-shock hazard when tested as specified in 4.4.7. Dimensions of the audiometer shall be approximately as follows: 13 inches wide by 7 inches high by 14 inches deep; it shall weigh approximately 12 pounds. An operational test shall be performed as indicated in 4.4.6.

3.4 Components. The audiometer shall consist essentially of the recorder, one pair of earphones, earphone covers, patch cords, patient operated hand switch, twelve fiber-tipped pens and one thousand audiogram charts. The electric power cord shall be a 3-wire, grounding type, having a minimum length of six feet, fitted with a Hospital Grade male plug conforming to American National Standards Institute (ANSI) Standard C73.11 and Underwriters' Laboratories (U.L.) Standard 498.

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3.5 Attenuation. The audiometer shall have the capability to automatically reset the attenuator to provide an output level well below a normal test subject's hearing threshold at the end of each test sequence at each test frequency. Rate of change of attenuation shall be 5 db. (nominal) per second, with the exception of the first excursion at each frequency, which shall be 10 db. (nominal) per second over the dynamic range of 100 db. Audiometer shall include a suitable instrument control to decrease tone level by 10 db. at each test frequency. Testing shall be as specified in 4.4.3.

3.6 Frequency. The audiometer shall enable selection of continuous or pulsed pure tone frequencies to include at least 500, 1000, 2000, 3000, 4000, 6000 and 8000 Hz. Frequency error shall not exceed ± 3 percent for any given frequency when tested as specified in 4.4.1.

3.7 Earphones and earphone covers. The earphones shall be of the dynamic moving-coil type. The earphones shall be individually calibrated at each of the seven discrete frequencies. Calibration shall be held to ± 2 db. for frequencies of 500 Hz through 3000 Hz, ± 3 db. for frequencies of 4000 Hz and 6000 Hz, and ± 4 db. for 8000 Hz. The sound pressure level (SPL) of the fundamental frequency shall be at least 30 db. above that of any harmonic. The root mean square (rms) weighted sound pressure, produced by the earphones due to all frequencies except the signal frequency and its harmonics, shall be either less than 0.001 dynes per square centimeter or at least 60 db. below the sound pressure, due to the signal frequency and its harmonics, at all designated frequencies and hearing loss levels. Any signal from the earphone not being used for test purposes shall have a sound pressure level at least 10 db. below the hearing threshold reference level, except that it need not be more than 70 db. below the signal from the active earphone. The headband shall exert a force of 1 to 2 kilograms laterally. The earphone bushing shall be constructed of sufficient strength to hold earphones properly positioned. Tests to determine conformance with the above requirements shall be conducted as specified in 4.4.2 and 4.4.4. The right earphone cover shall be marked in red and the left earphone cover shall be marked in blue for rapid identification.

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3.8 Hand-operated patient switch. Each audiometer shall be provided with a hand-operated patient switch, assembled to a rubber-insulated conductor at least three feet in length. The patient switch shall control the signal intensity across the hearing threshold so that the test tone will decrease when the button is depressed and will increase when the button is released. This allows the patient to track his threshold.

3.9 Power supply. Audiometer shall be designed to operate selectively at 115 and 230 volts, 50 and 60 Hz, alternating current. Audiometer shall have an internal voltage regulation circuit to compensate for ± 10 percent fluctuations of line voltage while maintaining the required calibration accuracy. The acoustic output of the earphones shall not vary through a total range of more than 2 db. as the line voltage is varied through the range required above. Testing shall be as specified in 4.4.5.

3.10 Recording system. The audiometer shall have a mechanical X-Y recorder in a printing system integral to the audiometer using a fiber tip pen for providing a graphic representation of the subject's threshold across all frequencies. The recording pen shall move in a continuous fashion once the hearing test commences. The recorder shall have a vertical axis corresponding to the intensity of the signal (from -10 db. to +90 db. Hearing Threshold Level - 1964 ISO Standards) and a horizontal axis corresponding to frequency and time. The recording pen shall be replaceable without any difficulty; replacement time shall not exceed 1 minute from the time the pen is removed from the X-Y drive mechanism of the unit as specified in 4.4.8.

3.11 Performance. Audiometer shall be a completely automatic machine. When the test is initiated, the audiometer shall automatically, without additional test operator intervention, present the test sequence in its entirety to both ears. The unit shall contain provisions for manually overriding the instrument at any time during a test so that a frequency may be skipped or repeated. The frequency shall automatically shift in proper register with the specific frequency on the audiogram card as the recording pen is moved manually. At the completion of the test, control shall be taken away from the subject under test and the signal shall be reduced to zero. Provision shall be made on the unit for either continuous or

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3.11 Performance (cont'd)

pulsed pure tones to be selected for presentation. The audiometer shall be designed so that the first test frequency of 500 Hz shall be directed to the left earphone. The 500 Hz signal shall be held for approximately 30 seconds. The signal shall then change automatically to 1000 Hz, and held at that value for 30 seconds. In succession, frequencies of 2000, 3000, 4000, 6000 and 8000 Hz, each sustained for 30 seconds, shall be presented to the left earphone. Following these seven presentations, the cycle shall continue with the 500 Hz signal directed to the right earphone through the seven frequencies automatically re-testing the right ear at 1000 Hz at the end of the sequence. Total test time for both ears shall be approximately 7-1/2 minutes. The sequence of timed operations which form a complete test operation shall be controlled by a timing motor which in turn operates cam drives or other suitable mechanisms. The cam drives or other mechanisms shall control resetting of the attenuator, control of mechanical devices for selection of appropriate frequency for specific points along the axis and control of switches for timed sequences. Counting of the subject's responses shall be by means of a binary counter. Circuitry shall include AND and OR gates for control of output signals to the counter and other components of the unit. The audiometer shall have: a TEST button or means for initiating the test or continuing the test after stop; a STOP button or means to interrupt signals and to return the intensity to minimum Hearing Threshold Level (HTL); a HOLD button or means to maintain current frequency or to permit manual frequency selection and to leave the subject in control of signal intensity. In addition, the audiometer shall have lamps to indicate "POWER ON" and "TEST" conditions to show if pulse is presented to subject under test.

3.12 Identification marking. Each audiometer shall be permanently and legibly marked with the following information:

Manufacturer's name or registered trademark
Model and serial number and electrical ratings (volt, amperes, watts and Hertz).

Identification marking shall be secured to unit.

3.13 Audiogram charts. One thousand audiogram charts in accordance with MIL-C-37187 shall be furnished with each audiometer (see 6.2)

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3.14 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer two preliminary copies of his proposed technical manual, for review and approval by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Material, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.15 Workmanship. Workmanship shall be first class throughout. Audiometers shall be free from defects which may impair their serviceability or may detract from their appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after such delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I.

TABLE I. Sampling for Examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Total (major and minor combined)	II	4.0

4.2.2 For tests. The performance test cited in 4.4.9 shall be performed on all audiometers. All other tests shall be conducted in accordance with Inspection Level S-1 of MIL-STD-105. The acceptance level shall be zero. Certification establishing proof that testing required in 4.4.6 has successfully been passed with an audiometer of the same design as that being supplied under current contract shall be considered acceptable.

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4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in Table II. Examination shall not be restricted to the classified possible defects listed below.

TABLE II. Classification of Defects

Categories and defects	
Major	
101	Any component part missing, broken or otherwise unusable.
102	Exposed edges not free of burrs or sharpness.
103	Any control dial or switch loose or does not operate smoothly.
104	Any indicator light does not operate properly.
105	Service data missing, not complete, or not as specified.
106	Headbands do not exert proper lateral pressure.
Minor	
201	Identification marking not complete, permanent or legible.
202	Identification marking not securely attached to unit.
203	Finish not free of chipped areas, cracks, scratches or roughness.

4.3.2 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

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TABLE III. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence, or method of application.

4.4 Tests. Tests shall be conducted to determine compliance with the requirements of this specification.

4.4.1 Accuracy of tone frequencies. The accuracy of the tone frequencies generated by the audiometer shall be determined by using a calibrated frequency counter.

4.4.2 Purity of tone. The sound pressure levels of the fundamental frequency and its harmonics shall be measured with the earphone coupled to a National Bureau of Standards Coupler 9-A. Measurements shall be made at all frequencies and at the 60 db. hearing-loss setting. The difference between the sound pressure levels of the fundamental and its harmonics shall be reported.

4.4.3 Accuracy of attenuator. Measurement of attenuator accuracy shall be made electrically at the input to the earphone terminated in a dummy load of 100 ohms.

4.4.4 Accuracy of sound pressure levels. The accuracy of the sound pressure levels at each frequency shall be determined with the earphone coupled to a National Bureau of Standards 9A Coupler. The accuracy of the sound pressure levels at each frequency after a minimum of one hour of operation shall also be determined. The sound pressure levels obtained shall conform to requirements as indicated in America National Standards Institute (ANSI) S3.6.

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4.4.5 Stability with respect to variations in supply voltage. The audiometer shall be tested over the full range of supply voltages to determine compliance with requirements. The voltage during test shall be varied from 105 to 125 VAC for 115 volt, 50-60 Hertz operation. Tests for compliance with requirements shall be made at a 70 db. hearing threshold level setting and at a frequency of 1000 cps.

4.4.6 Noise during test.

4.4.6.1 Sound from test earphones. The rms weighted sound pressure, due to all frequencies except the signal frequency and its harmonics, shall be determined with the earphone coupled to a National Bureau of Standards 9A Coupler.

4.4.6.2 Sound from second earphone. The sound pressure level in the earphone not being used for test purposes shall be measured to determine compliance with requirements. This test shall be performed by measuring the electrical signal across a dummy load of 100 ohms which simulates the impedance of the earphone.

4.4.6.3 Other unwanted sound. A subject having normal hearing shall be located in the normal test position wearing a pair of disconnected earphones. The electrical output of the audiometer shall be absorbed in a 100 ohm resistor while the test is being performed. The audiometer shall be operated through a normal test cycle at a hearing loss setting of 50 db. The subject shall listen for any sound emanating from the audiometer which could furnish a clue that could influence the hearing test results. As an alternate the test may be performed as indicated in American National Standards Institute (ANSI) S3.6. This test is not intended to apply to sounds from a mechanism such as the hand-operated switch, since the subject knows when he presses this switch.

4.4.7 Shock hazard. The audiometer shall be tested to determine if an electrical shock hazard exists. A shock hazard shall be considered to exist at an exposed part if the open circuit potential to ground is more than 30 volts rms and the current through a 1500 ohm resistor is more than 5 milliamperes.

4.4.8 Recording pen replacement. Using a suitable time measuring device such as a stop watch, mechanical or electrical counter, etc., the time duration involved from the removal of the original recording pen to the replacement with another recording pen shall be recorded. The instructions in the operating manual supplied with the unit shall be followed.

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4.4.9 Performance test. Each audiometer shall be operated and tested to establish that it performs all functions and operations required in 3.11.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. The one thousand charts shall be packaged in a snug fitting box constructed in accordance with PPP-B-566 or PPP-B-676. The twelve pens shall be individually wrapped in neutral tissue or other suitable non-corrosive material and shall be individually or collectively packaged in a suitable size and design. The earphones, earphone covers, patch cords, and handswitch shall be individually wrapped in neutral tissue or other suitable material and individually or collectively packaged in a box of appropriate size and design constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, class domestic. The individually or collectively packaged items may be consolidated within a box constructed in accordance with PPP-B-636, class domestic. The audiometer along with the packaged accessories shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Each unit shall be packed in an exterior container designed for a type 1 load and constructed in accordance with PPP-B-601, overseas type or PPP-B-636, class weather-resistant, variety DW. Wood boxes shall be lined with fiberboard or heavy weight kraft paper to prevent chafing of the waterproof barrier. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.2 Level B. Each unit shall be packed as specified in 5.2.1 except that the exterior containers shall be domestic types and strapping shall not be required.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual packages. Individual packages shall be marked to identify contents and quantity contained.

5.3.2 Unit packages. The unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-536-5056	AUDIOMETER, AUTOMATIC, 115-230 volt, 50-60 Hz, AC, Solid State

6.2 Intended use. The audiometer covered by this specification is intended to be used with National Stock No. 6515-00-536-5059 Chart, Recording Instrument, 1000s. This chart is for use with audiometers calibrated to American National Standard (International Standards Organization) reference levels.

6.3 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

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6.4 This specification does not include all types, classes, grades and sizes, of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1922

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-A-37192(DSA-DM) AUDIOMETER, AUTOMATIC

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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